

10 (**long**) The temperature in a room is given by

$$T = 20 + A \sin [(xy + x + y + 1 - z)/d]$$

where T is the temperature in degrees Celsius, (x, y, z) are the Cartesian coordinates in metres, the amplitude $A = 1^\circ\text{C}$ and the scale factor $d = 1\text{ m}$.

(a) Find an expression for ∇T . What are the units of ∇T ? [4]

(b) Determine the equation of the plane that is tangent to the constant-temperature surface at the point $(1, 0, 2)$. [6]

(c) Sketch contours of constant temperature in the plane $z = 0$. [6]

(d) A bee is flying along a spiral trajectory. Its position is given by $B(\cos 2t, \sin 2t, 2t/\pi)$ metres, where t is the time in seconds and the amplitude $B = 1\text{ m}$. Find the rate of change of temperature experienced by the bee as it passes through the point $(1, 0, 2)$. [8]

(e) A wasp flies at constant speed 1 ms^{-1} through the point $(1, 0, 4)$, along a straight-line trajectory in the direction $[1\ 0\ 1]^t$. Sketch a graph showing the temperature experienced by the wasp as a function of time. [6]